

Chapter 7

Advanced Encryption Standard (AES)

進階加密標準

- ❑ To review a short history of AES
- ❑ To define the basic structure of AES
- ❑ To define the transformations used by AES
- ❑ To define the key expansion process
- ❑ To discuss different implementations

7-1 INTRODUCTION

*The Advanced Encryption Standard (AES) is a symmetric-key block cipher published by the **National Institute of Standards and Technology (NIST)** in December 2001.* 是美國國家標準技術局NIST於2001年12月所發布的對稱式區塊加密法。

AES is to replace DES

Topics discussed in this section:

7.1.1 History

7.1.2 Criteria

7.1.3 Rounds

7.1.4 Data Units

7.1.5 Structure of Each Round



7.1.2 *Criteria* 評選準則

The criteria defined by NIST for selecting AES fall into three areas:

- 1. Security*
- 2. Cost*
- 3. Implementation.*

7.1.3 Rounds 回合數

*AES is a **non-Feistel cipher** that encrypts and decrypts a **data block of 128 bits (16 bytes)**.*

*It uses **10, 12, or 14 rounds**.*

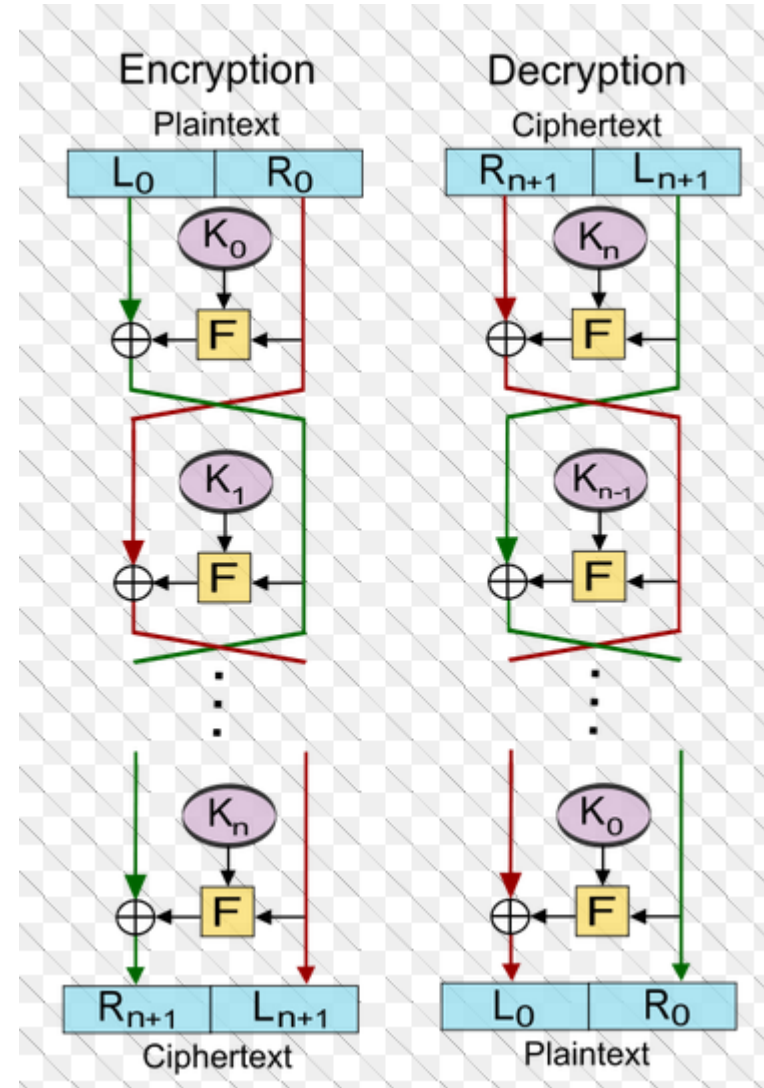
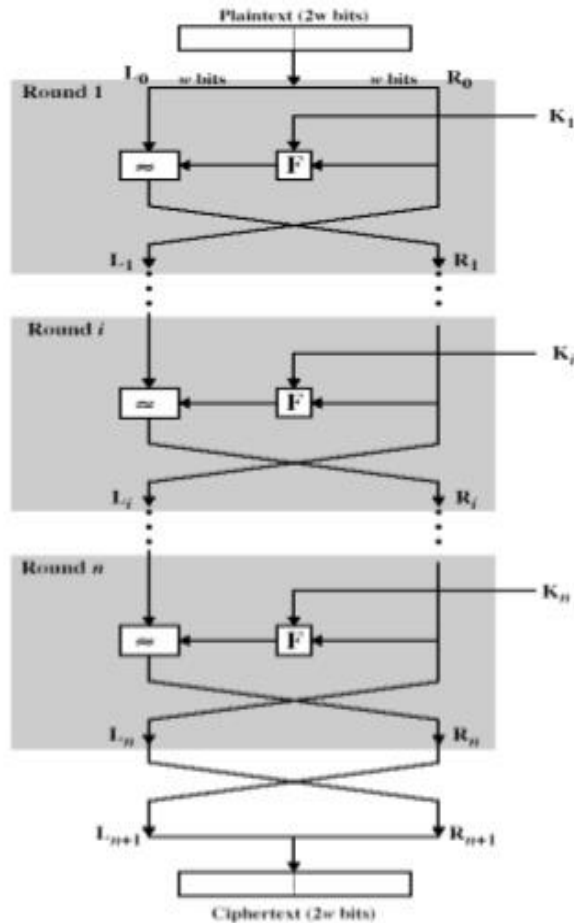
*The **key size**, which can **be 128, 192, or 256 bits**, depends on the number of rounds.*

Note

AES資料區塊的長度為 **128 位元**

總共定義了三個版本，分別有十、十二和十四個回合的運算，每個版本使用的金鑰長度也都有所不同（分別是 **128**、**192** 或 **256 位元**），但是所有的回合金鑰都是 **128 位元**。

Feistel Cipher Structure



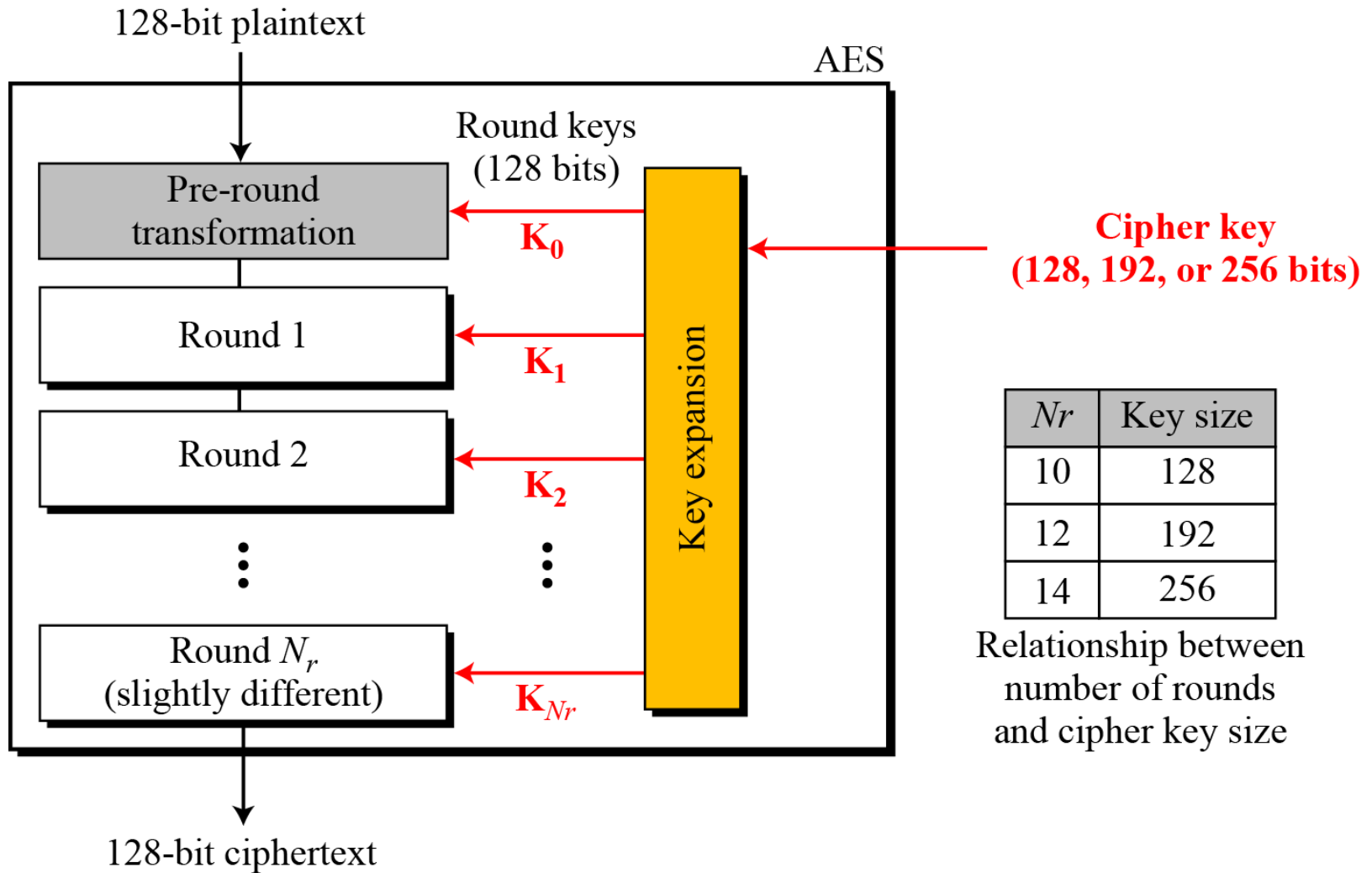
29

DES is Feistel Cipher structure

3DES: 密文 = $E_{K_3}(D_{K_2}(E_{K_1}(\text{明文})))$, 明文 = $D_{K_1}(E_{K_2}(D_{K_3}(\text{密文})))$

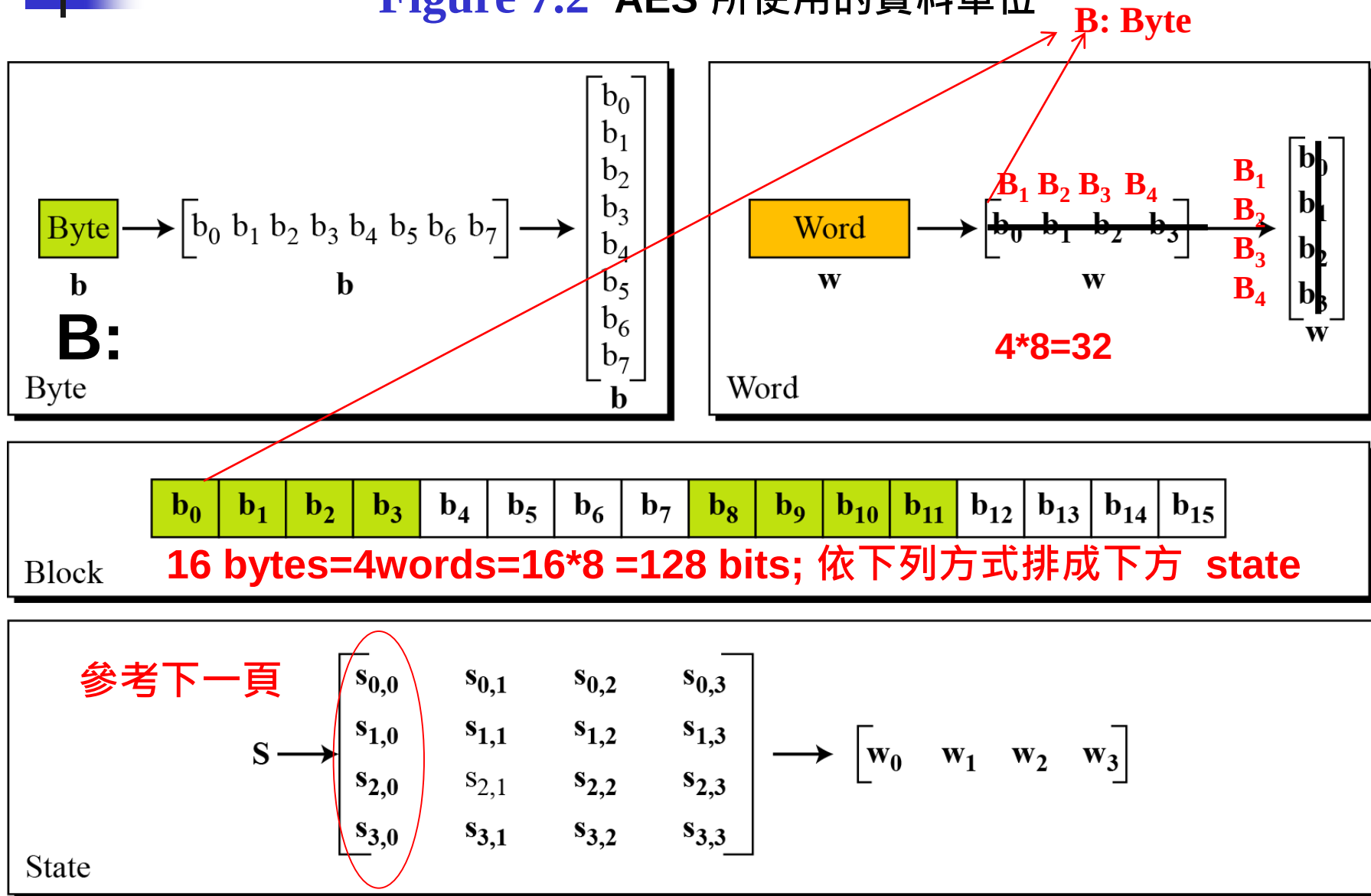
7.1.3 AES 加密法的設計概念 e

Figure 7.1 *General design of AES encryption cipher*



7.1.4 Data Units.

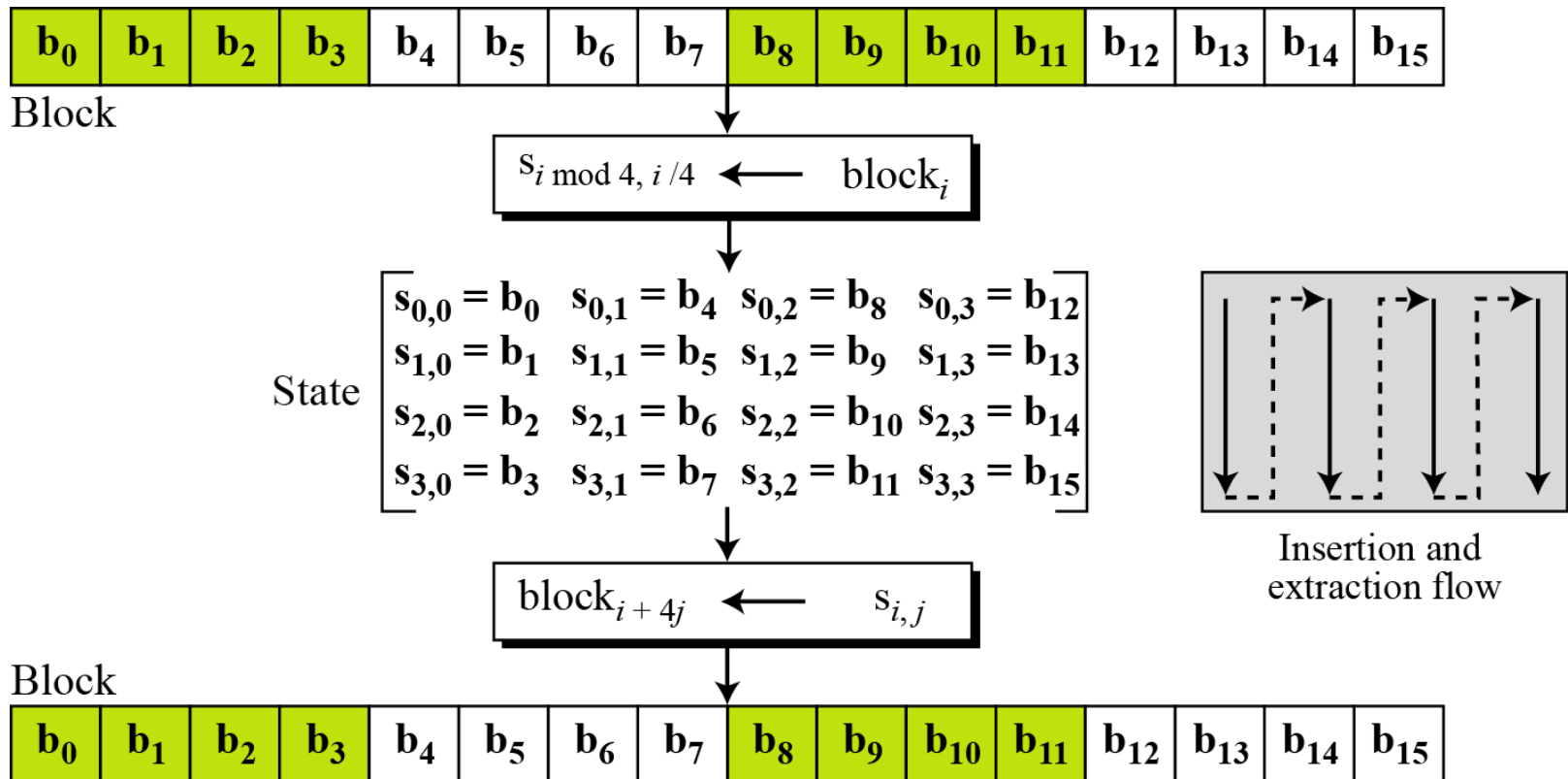
Figure 7.2 AES 所使用的資料單位



7.1.4 區塊與狀態之間的轉換

Figure 7.3 *Block-to-state and state-to-block transformation*

b 改成 B



7.1.4 將明文轉成state

Example 7.1

State : (16 bytes)=4 words= 128bits

Figure 7.4 Changing plaintext to state

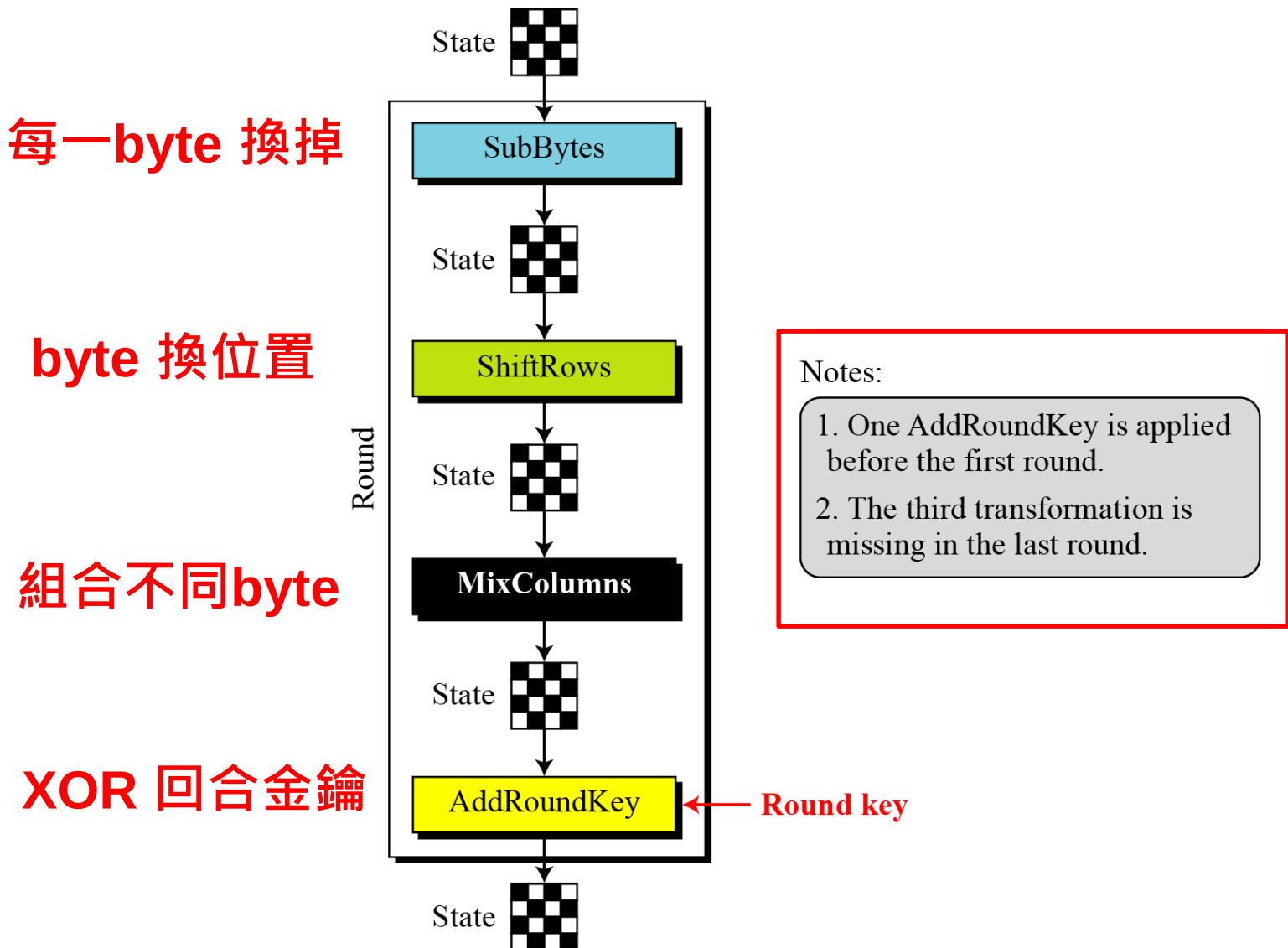
Text	A	E	S	U	S	E	S	A	M	A	T	R	I	X	Z	Z
Hexadecimal	00	04	12	14	12	04	12	00	0C	00	13	11	08	23	19	19

00	12	0C	08
04	04	00	23
12	12	13	19
14	00	11	19

State

7.1.5 加密端每個回合的結構

Figure 7.5 Structure of each round at the encryption site



7.2.1 Substitution

AES, like DES, uses substitution. AES uses two invertible transformations. AES使用兩個可逆的轉換加密端的SubBytes 與解密端的InvSubBytes 程序。

SubBytes

*The first transformation, **SubBytes**, is used at the encryption site. To substitute a byte, we interpret the byte as two hexadecimal digits.*

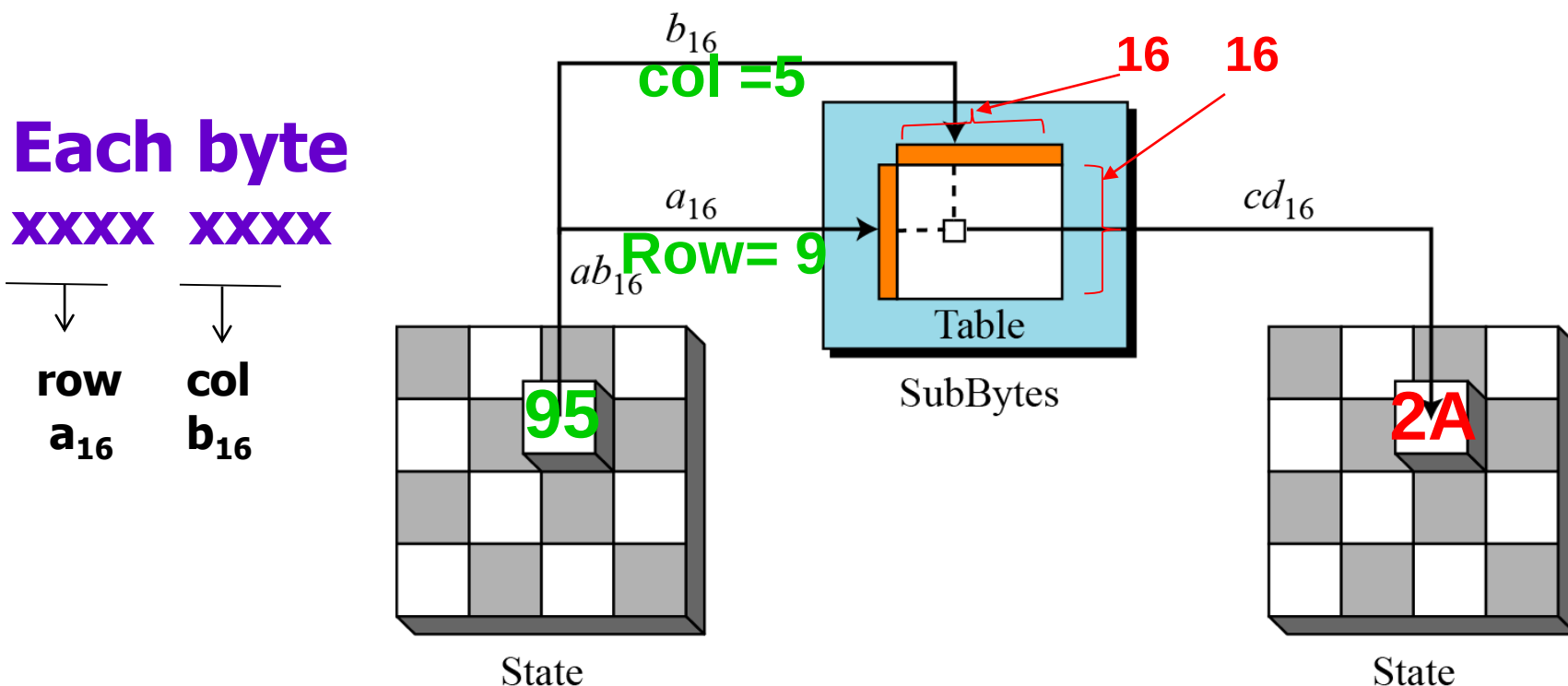
Note

The SubBytes operation involves 16 independent byte-to-byte transformations.

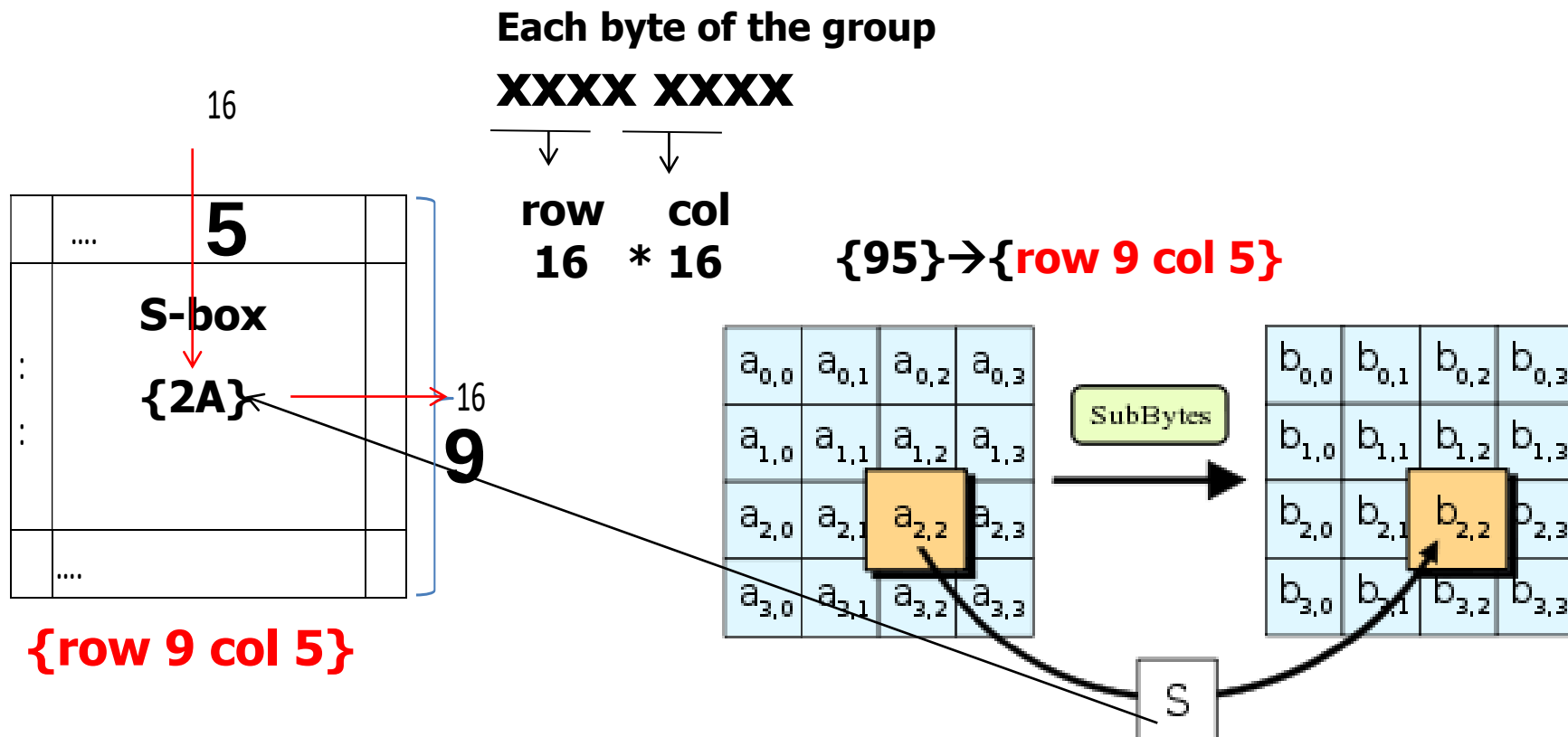
7.2.1 Continue

在取代一個位元組的時候，首先將 位元組 (8 bits) 表示成 2個 十六進位的數字

Figure 7.6 *SubBytes transformation*



Byte Substitution



7.2.1 SubBytes transformation Table (s-box)

Table 7.1 SubBytes transformation table

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	63	7C	77	7B	F2	6B	6F	C5	30	01	67	2B	FE	57	AB	76
1	CA	82	C9	7D	FA	59	47	F0	AD	D4	A2	AF	9C	A4	72	C0
2	B7	FD	93	26	36	3F	F7	CC	34	A5	E5	F1	71	D8	31	15
3	04	C7	23	C3	18	96	05	9A	07	12	80	E2	EB	27	B2	75
4	09	83	2C	1A	1B	6E	5A	A0	52	3B	D6	B3	29	E3	2F	84
5	53	D1	00	ED	20	FC	B1	5B	6A	CB	BE	39	4A	4C	58	CF
6	D0	EF	AA	FB	43	4D	33	85	45	F9	02	7F	50	3C	9F	A8

col 5

row 9

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
7	51	A3	40	8F	92	9D	38	F5	BC	B6	DA	21	10	FF	F3	D2
8	CD	0C	13	EC	5F	97	44	17	C4	A7	7E	3D	64	5D	19	73
9	60	81	4F	DC	28	2A	90	88	46	EE	B8	14	DE	5E	0B	DB
A	E0	32	3A	0A	49	06	24	5C	C2	D3	AC	62	91	95	E4	79
B	E7	CB	37	6D	8D	D5	4E	A9	6C	56	F4	EA	65	7A	AE	08
C	BA	78	25	2E	1C	A6	B4	C6	E8	DD	74	1F	4B	BD	8B	8A
D	70	3E	B5	66	48	03	F6	0E	61	35	57	B9	86	C1	1D	9E
E	E1	F8	98	11	69	D9	8E	94	9B	1E	87	E9	CE	55	28	DF
F	8C	A1	89	0D	BF	E6	42	68	41	99	2D	0F	B0	54	BB	16

95 → 2A
0C → FE

7.2.1 *InvSubBytes*

Table 7.2 *InvSubBytes* transformation table

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	52	09	6A	D5	30	36	A5	38	BF	40	A3	9E	81	F3	D7	FB
1	7C	E3	39	82	9B	2F	FF	87	34	8E	43	44	C4	DE	E9	CB
2	54	7B	94	32	A6	C2	23	3D	EE	4C	95	0B	42	FA	C3	4E
3	08	2E	A1	66	28	D9	24	B2	76	5B	A2	49	6D	8B	D1	25
4	72	F8	F6	64	86	68	98	16	D4	A4	5C	CC	5D	65	B6	92
5	6C	70	48	50	FD	ED	B9	DA	5E	15	46	57	A7	8D	9D	84
6	90	D8	AB	00	8C	BC	D3	0A	F7	E4	58	05	B8	B3	45	06
7	D0	2C	1E	8F	CA	3F	0F	02	C1	AF	BD	03	01	13	8A	6B
8	3A	91	11	41	4F	67	DC	EA	97	F2	CF	CE	F0	B4	E6	73
9	96	AC	74	22	E7	AD	35	85	E2	F9	37	E8	1C	75	DF	6E
A	47	F1	1A	71	1D	29	C5	89	6F	B7	62	0E	AA	18	BE	1B
B	FC	56	3E	4B	C6	D2	79	20	9A	DB	C0	FE	78	CD	5A	F4
C	1F	DD	A8	33	88	07	C7	31	B1	12	10	59	27	80	EC	5F
D	60	51	7F	A9	19	B5	4A	0D	2D	E5	7A	9F	93	C9	9C	EF
E	A0	E0	3B	4D	AE	2A	F5	B0	C8	EB	BB	3C	83	53	99	61
F	17	2B	04	7E	BA	77	D6	26	E1	69	14	63	55	21	0C	7D

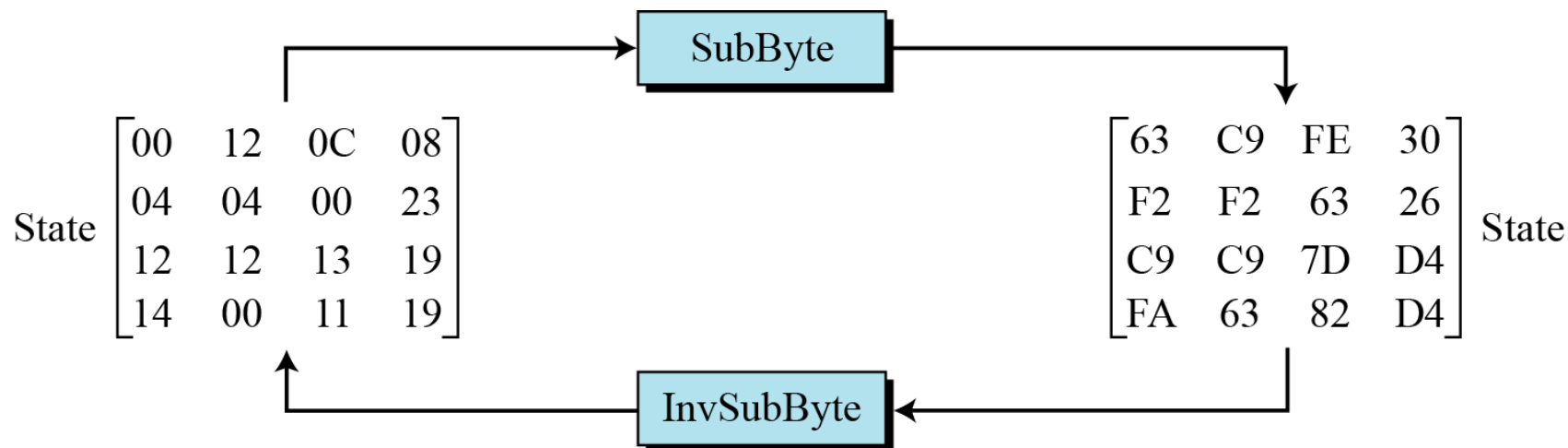
2A → 95
FE → 0C

7.2.1 Continue

Example 7.2

Figure 7.7展示一組狀態如何使用 **SubBytes** 進行轉換，該圖同樣也展示出 **InvSubBytes** 轉換可產生原始的狀態。
Note that if the two bytes have the same values, their transformation is also the same.

Figure 7.7 *SubBytes transformation for Example 7.2*



SubBytes 與 InvSubBytes 轉換互為反向。

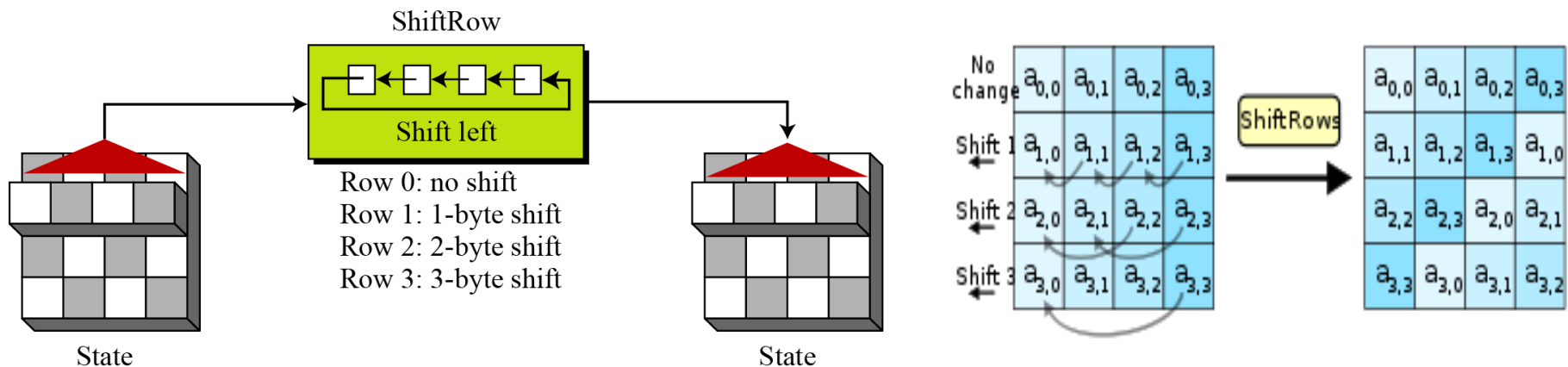
7.2.2 Permutation 排列

ShiftRows

In a round of the encryption, another transformation, called ShiftRows, *permutes(or shifts) the bytes*.

位移是由右向左

Figure 7.9 ShiftRows transformation

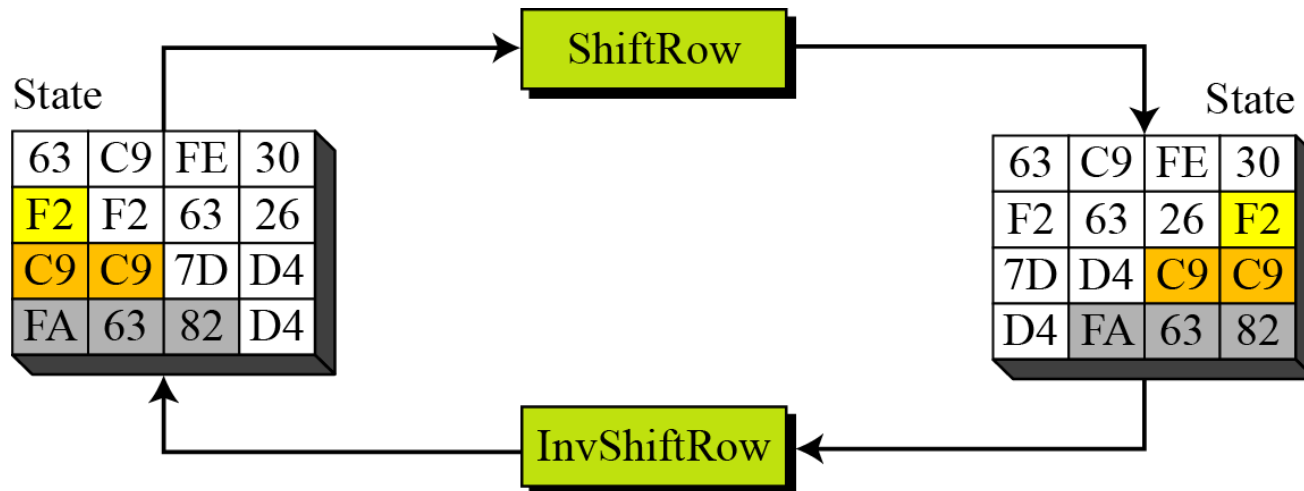


7.2.2 Continue– 参考

Example 7.4

Figure 7.10 shows how a state is transformed using ShiftRows transformation. The figure also shows that InvShiftRows transformation creates the original state.

Figure 7.10 *ShiftRows transformation in Example 7.4*

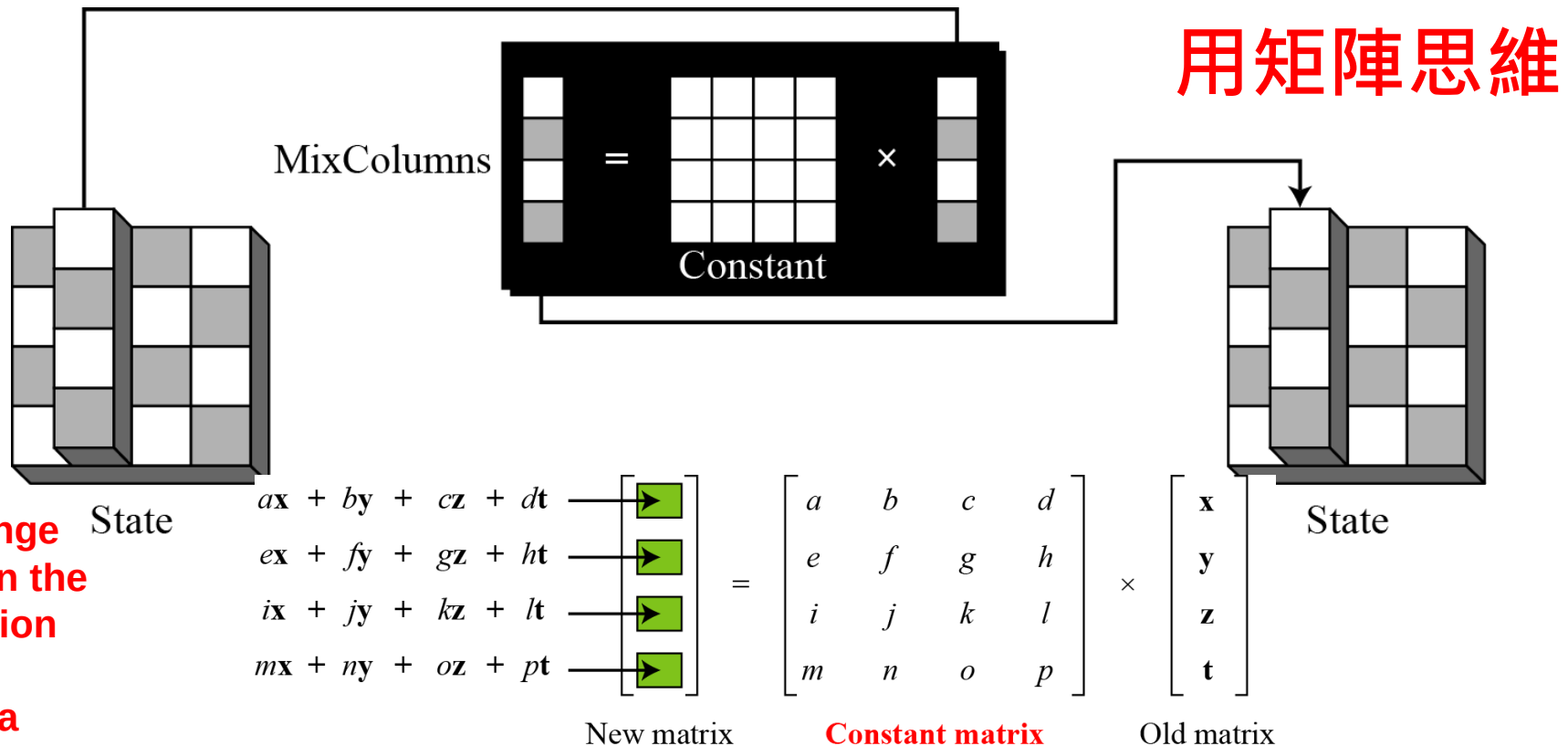


7.2.3 MixColumns Transformation

MixColumns

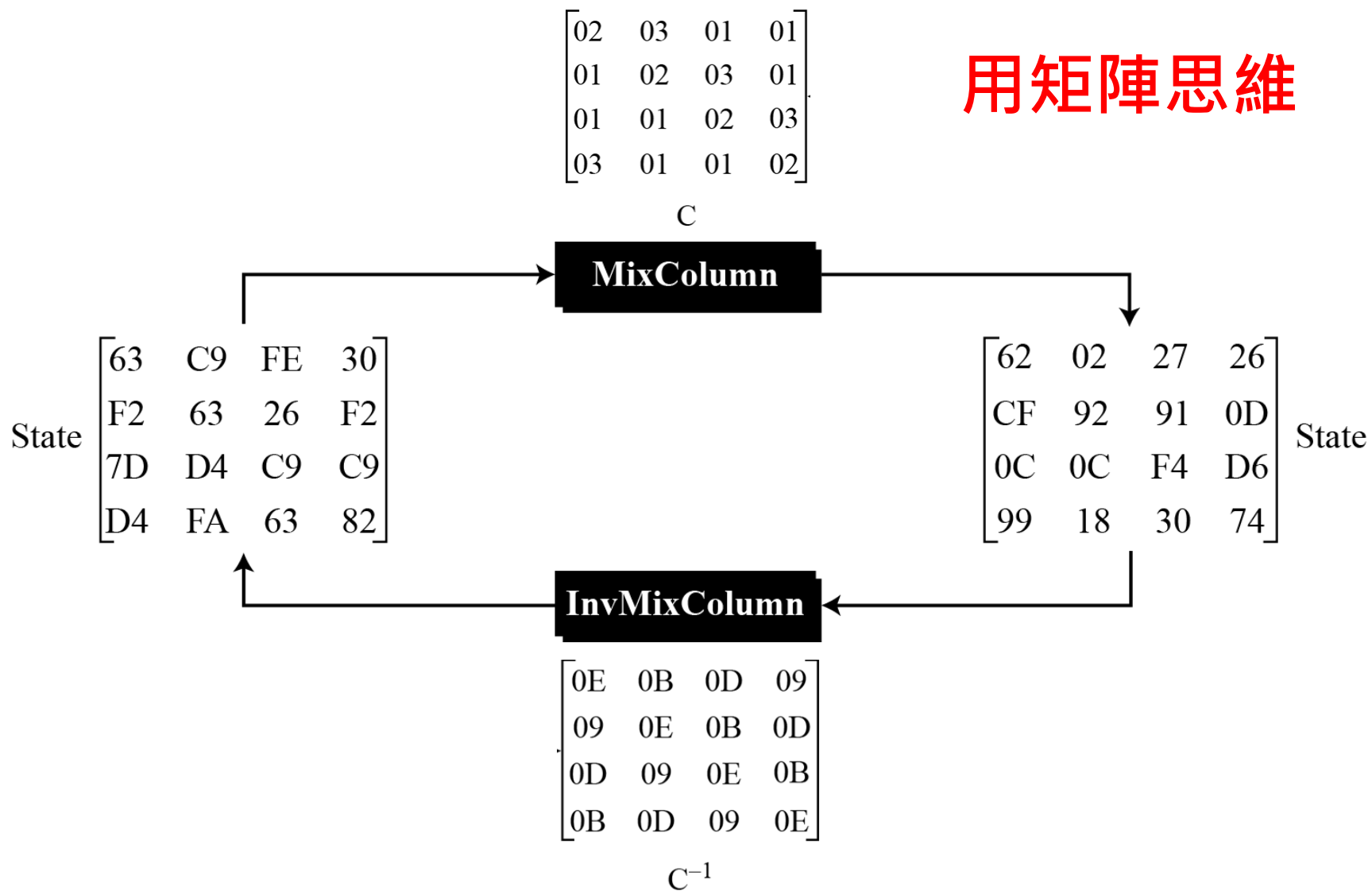
The MixColumns transformation operates **at the column level**; it **transforms each column of the state to a new column**.

Figure 7.13 MixColumns transformation



7.2.3 Example 7.5– 參考

Figure 7.14 *The MixColumns transformation in Example 7.5*



7.2.4 Key Adding

AddRoundKey

XOR state with 128-bits of the round key

AddRoundKey proceeds **one column** at a time.

AddRoundKey 將回合金鑰加入狀態矩陣行；

AddRoundKey 使用矩陣加法。

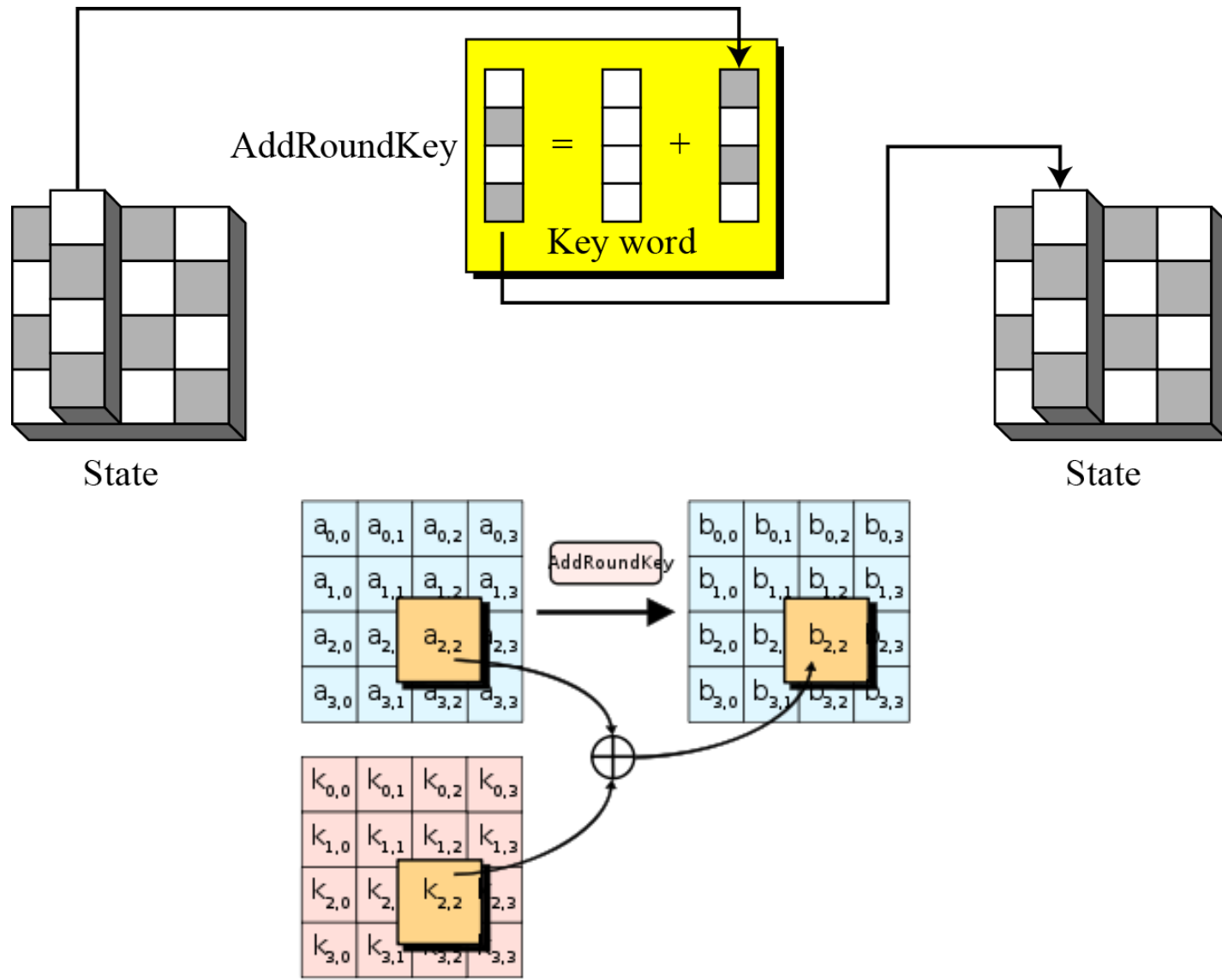
Note

inverse for decryption is identical since XOR is own inverse,

The AddRoundKey transformation is the inverse of itself.

7.2.4 Continue

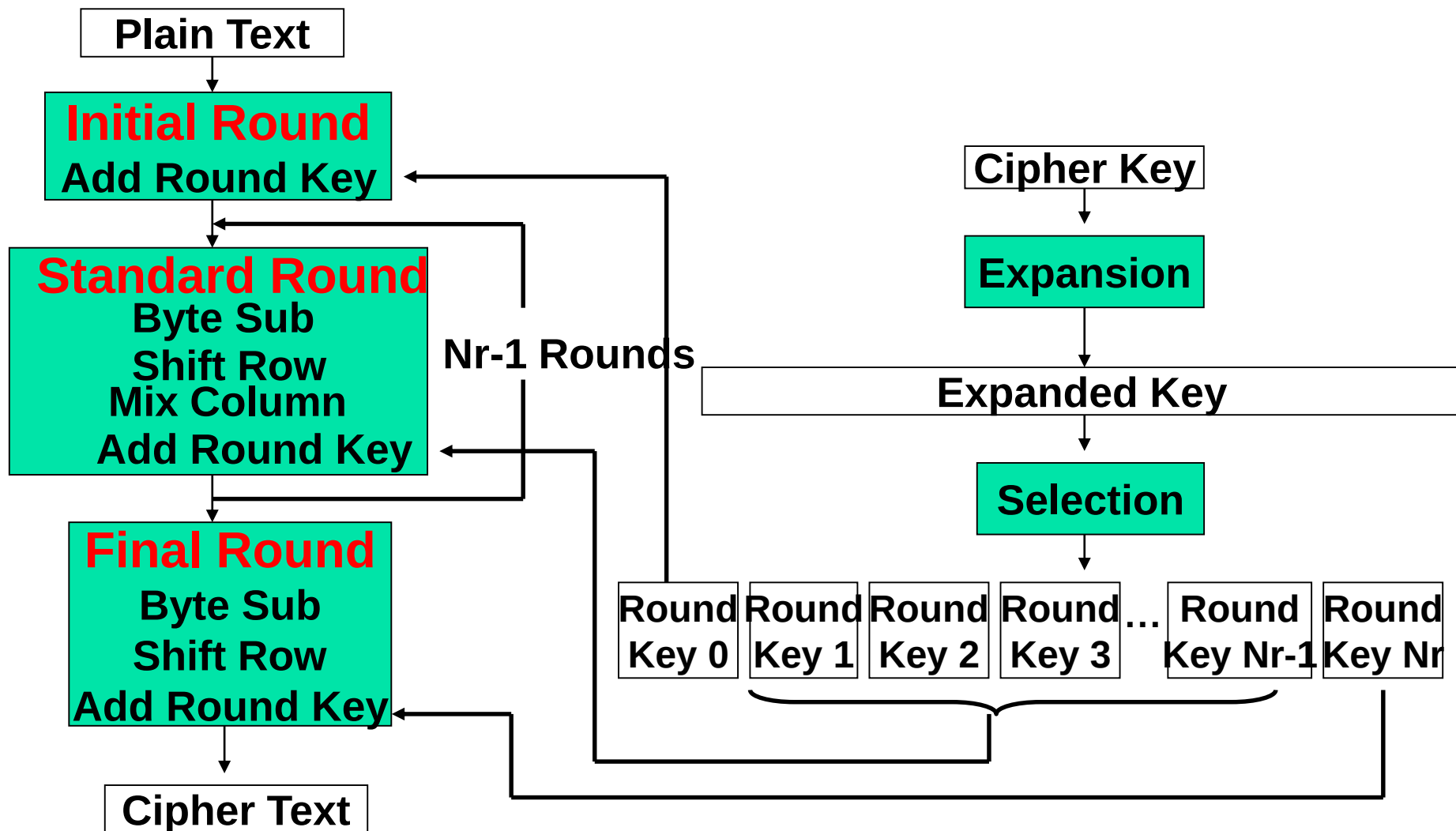
Figure 7.15 *AddRoundKey transformation*





Rijndael 的加密演算法

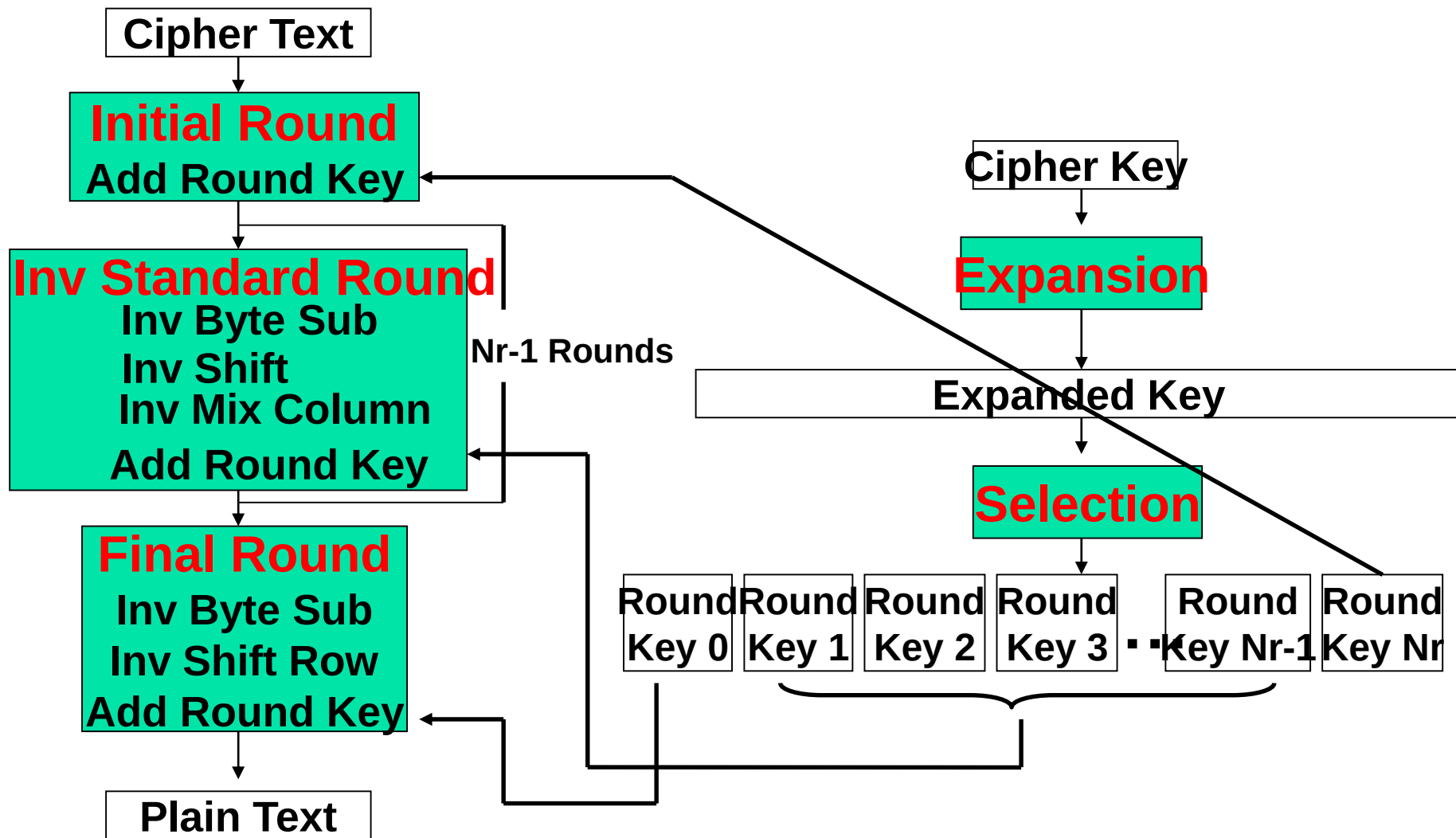
新一代密碼系統





Rijndael的解密演算法

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Videos



Joan Daemen

Vincent Rijmen

■ Video

- AES Rijndael Cipher – Visualization
<http://www.youtube.com/watch?v=mlzxpkdXP58>
- AES (Rijndael) Explained Part 1~5 (w voices)-
<http://www.youtube.com/watch?v=ySq88y0e8u4&feature=related>
- VB 2010 [Tutorial] How To Encrypt/Decrypt File Contents Using Custom DLL
<http://www.youtube.com/watch?v=VmmRO5szjvA&feature=related>
- Modes of AES Encryption - Part 1 -
<http://www.youtube.com/watch?v=BGHCcDvJTek&feature=related>

Videos



- Modes of AES Encryption - Part 1 -
<http://www.youtube.com/watch?v=BGHCcDvJTEk&feature=related>
- Rijmen-Daemen in Belgium
<http://www.youtube.com/watch?v=Fk1jGo8uzcY>
- Vincent Rijmen - IAIK - Graz University of Technology

http://www.iaik.tugraz.at/content/about_iaik/people/rijmen_vincent/

